AN ETALE MORPHISM TO A CLOSED SUBSCHEME CAN BE COVERED BY RESTRICTIONS OF ETALE MORPHISMS TO THE AMBIENT SCHEME

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the reduction to the affine case is justified in the counted span and the affine case is attributed explicitly.

Editorial note. The label lemma-closed-immersion-almost-essentially-surjective occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference occurring inside the counted spans points to a chapter of the Project that is not part of this card's source tree; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Closed immersion i : Z -> X and the notation U_Z for the base change of an etale X-scheme. Same chapter, the notation U_Z = U x_X Z for U etale over X. Supplies the objects of the required Zariski covering of V and the morphisms U_{i,Z} -> V.
- Zariski localization of the problem. Same chapter, the standard reduction allowing a statement about Zariski coverings to be checked after Zariski localization. Supplies the reduction to the case where X and V are affine.
- Lifting an etale morphism along a closed immersion over an affine base. Algebra, the result that an etale morphism to a closed subscheme of an affine scheme is Zariski locally the restriction of an etale morphism to the ambient scheme. Supplies the affine case, which is the whole content after the reduction.

Source reference. Original lemma 04D1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6633–6633. Cross-chapter navigation pointer invoked at proof line 6633, where the author reduces the affine case to the corresponding statement in the chapter Algebra. The target chapter is staged in this repository but not in this card's source tree, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $i : Z \to X$ be a closed immersion of schemes. Let $V \to Z$ be an étale morphism of schemes. There exist étale morphisms $U_i \to X$ and morphisms $U_{i,Z} \to V$ such that $\{U_{i,Z} \to V\}$ is a Zariski covering of V.*

Proof. Since we only have to find a Zariski covering of V consisting of schemes of the form U_Z with U étale over X, we may Zariski localize on X and V. Hence we may assume X and V affine. In the affine case this is Algebra, Lemma 04D1. □